D130 | A network is not an app moat

PRIVATE FICTIONAL analysis. No real payment data, financial product, merchant experiment, app/account change, contact, data-use permission or spend.

Separate layers	NPCI operates interoperable rails; PSP/application and bank roles differ. The same bank can hold more than one role. A TPAP is not automatically the remitter bank or exclusive owner of merchant reach. Shared reach does not prove identical reliability, service, trust or distribution.
Completion experiment	Comparable payer cohorts: treatment 60% to 66%, control 61% to 63%, same window. DiD is 4 percentage points, not 3-point final gap or 6-point treatment change. Toy random assignment is stipulated, not observed; spillovers, changing mix and clustered dependencies need design checks.
Economic model	Unrelated monthly toy:10000 transactions, revenue 0.30/cost 0.50 rupees; 100 subscribers revenue 30/cost 5; fixed 1000. Net minus 500. These numbers are NOT UPI MDR, payer charges, current law or any company actual figures.
Multihoming	Second app 30 minutes/month at 200 rupees/hour plus 20 administration and no provider fee gives 120/month. Using two apps is not switching or abandoning one. No provider fee does not make time and administration zero.
Policy boundary	Dated government summary and NPCI circular directory are not a current fee calculator. Selected September 2026 consolidated circular states pricing circulars remain applicable; it does not replace missing operative pricing instruments. No blanket zero-MDR or selective-rate rule is applied here.
Evidence boundary	Written selected official/VC sources refreshed 13 November 2026; no full diagram/AV coverage or independent causal validation. Actual learner/reviewer/delayed transfer blank. No data monetization entitlement, legal/account clearance or overall 10/10 claim.

Inputs | Separate denominators and units

Fractions 0..1 for completion. Monthly rupees and counts for unrelated economics; minutes/month and rupees/hour for time. Yellow controls are fictional, not a payment rule.

Treatment before completion fraction	0.60
Treatment after completion fraction	0.66
Control before completion fraction	0.61
Control after completion fraction	0.63

Monthly transaction count	10000.00
Revenue per transaction, INR	0.30
Cost per transaction, INR	0.50
Subscriber count	100.00
Revenue per subscriber/month, INR	30.00
Cost per subscriber/month, INR	5.00
Monthly fixed cost, INR	1000.00

Second-app minutes/month	30.00
Value of time, INR/hour	200.00
Administration, INR/month	20.00
Provider fee, INR/month, fictional	0.00

Completion fractions valid?	1.000000
Economics inputs valid?	1.000000
Multihoming inputs valid?	1.000000

Transaction count must be a positive integer for average break-even revenue; subscribers may be zero. Nonnegative inputs do not require positive contribution. Blank/text/negative/out-of-range inputs hold only their own model.

Roles | Operator, app and bank are not synonyms

Selected NPCI overview/participation/authentication clauses, not a full payment-system specification or architecture certification.

NPCI	Owns/operates UPI and provides routing, processing, clearing and settlement framework to participating institutions. Does not imply that each app owns the interoperable network or all merchant/customer relationships.
PSP and application	PSP supplies payment-service participation and UPI ID role; app provides user interface and services. TPAP participates through PSP arrangements. More than one role may sit with one institution. Do not collapse payer PSP, remitter bank, TPAP and NPCI into one party.
Remitter and beneficiary	Payer is sending user; remitter bank holds/debits linked payer account and authenticates PIN/account authorization. Payee receives; beneficiary bank holds/credits receiving account. Payee PSP/beneficiary bank conveys beneficiary name. This is a simplified role map, not every exception/product.
Different authentication roles	Consolidated clauses 3.4/3.5 distinguish payer PSP first-factor/device-binding responsibility from remitter-bank PIN/account authentication. Do not collect PINs or use a classroom model to change account access.
Interoperability	Shared reach lowers exclusive access as an app advantage. Reliability, support, distribution and trust can still differ. One shared merchant network does not make every app equally useful or equally defensible.
Consent is separate	Selected data/privacy clauses and app functionality do not authorize our data collection, processing, disclosure or adjacent monetization. No real payer/merchant record, signup or data export is used.

Completion | Four cells, not one growth rate

Randomized availability assignment and comparable cohorts are STIPULATED only. No sample sizes, standard errors, compliance, clustering or real experiment supplied.

Treatment change, percentage points	6.000000
Control change, percentage points	2.000000
Difference-in-differences, percentage points	4.000000
Final treatment-control gap, points	3.000000
Control-trend counterfactual fraction	0.620000

DiD relative to treatment baseline, fraction	0.066667

Baseline 6-2=4 percentage points; final gap 3; trend counterfactual.62; relative lift.0666667, not 4%. Algebra alone does not prove a causal network effect. Control using treatment merchants creates spillovers; assignment/inference units, cohort mix, window, access and failure definitions need checking. No sample size means no precision or significance claim.

Economics | Contribution is not volume

Unrelated fictional monthly model. All rates are rupees per stated unit. Subscription contribution counts separately; no actual UPI fees, subsidy, customer data revenue or double counting assumed.

Transaction margin, INR/transaction	-0.200000
Monthly transaction contribution, INR	-2000.000000
Subscription margin, INR/subscriber/month	25.000000
Monthly subscription contribution, INR	2500.000000
Net after fixed costs, INR/month	-500.000000

Break-even revenue per transaction, INR	0.350000
Continuous subscriber break-even, fixed volume	120.000000
Whole subscriber threshold, fixed volume	120.000000

Baseline transaction margin-.20, contribution-2000; subscriptions 25 each =>2500; net-500. Fixed 100 subscribers needs transaction revenue.35 for net 0. At fixed 10000 transactions/25 subscription margin need 120 subscribers. Nonpositive subscription margin has no positive-margin threshold here; more loss-making transactions can worsen net. Capacity, taxes and acquisition/retention costs not supplied.

Multihome | Free to install is not costless

Separate monthly second-app burden. Time is valued at a fictional opportunity cost; not an invoice, vendor fee, switching cost or direction to create an account.

Hours/month	0.500000
Time cost, INR/month	100.000000
Total second-app cost, INR/month	120.000000
Annual arithmetic equivalent, INR	1440.000000

Baseline .5 hour*200=100 plus 20=120/month, 1440/year only if unchanged for 12 months. This is ongoing multihoming, not one-off migration or account closure. Higher second-app burden can matter, but does not establish actual exclusivity or permission to impair interoperability.

Mechanisms | Name who gains and how

Hypotheses to test, not claims that every participant produces positive network value. More volume can reflect marketing, macro demand or subsidies.

Cross-side merchant to payer	More usable accepting merchants can improve payer coverage. Measure success/access for comparable payer tasks, at fixed service quality. Shared/interoperable reach may spread gains across apps, not exclusively to the adding app.
Cross-side payer to merchant	More active relevant payers can improve reachable demand, conditional on actual purchases, costs and access. Count retention/contribution, not registered accounts alone; merchant congestion/fraud/support burden may offset benefits.
Same-side is not automatic	Additional payers might improve peer-payment reach if reachable and useful; additional merchants may create discovery variety or competition. Identify beneficiary and mechanism rather than labeling every increase as positive same-side effects.
Negative effects and saturation	Congestion, fraud, irrelevant inventory and support load can lower quality. Track failures, latency, harm and satisfaction. Historical five-minute rideshare or seven-person lending cutoffs in a 16 z are examples, not universal thresholds.
Moat and causality	Interoperability, network overlap and multihoming can weaken exclusive reach. Reliability, trust and service may differ. Network labels and retention correlations alone do not identify causal defensibility or permanent winner-take-all.
No actual experiment	Original randomized toy is paper analysis only. No payer/merchant contact, tracking, dark pattern, market intervention, privacy change or financial action is authorized.

Policy | Dates do not settle applicability

No operative fee/cap calculator. Keep a policy ledger separate from synthetic economics. Current legal/product answer remains open pending exact instruments and applicability.

NPCI directory 13 Nov 2026	OC237 and 238 list Please get in touch with your bank. A directory title is not operative pricing terms or a concentration-cap instrument. We do not contact a bank or accept terms for this exercise.
Consolidated 18 Sep 2026	Version 1.0 states consolidation of operating circulars through 31 July 2026. Clause 1.5 preserves pricing circulars; footnote says no alteration of fee, charge, effective date/compliance timeline/technical requirements. Selected introductory/roles sections read, not full 53 pages or all annexures.
Government 15 Sep 2026	PIB summary reports free P2P and scoped P2M MDR framework, including selective merchant rates/exemptions and no customer MDR. This is a dated government summary, not our verified operative pricing instrument or a universal zero-MDR conclusion. Rates not copied into the economic model.
Missing operative bridge	Need exact notification/circular text, amendments, effective date, account/instrument/product, merchant category, exceptions, fee bearer and binding implementation. Consolidation cutoff precedes summary, so cannot treat it as later replacement pricing law. No current concentration cap asserted.
Scope tension to resolve	Selected consolidated merchant clause 7.5 discusses surcharge/upfront convenience disclosure; PIB summary prohibits app platform/hidden fees. Different entity, product, fee definitions and dates may matter. Do not average, merge or turn either statement into blanket customer-fee advice.
Decision held	No real fee quote, customer advice, merchant classification, account change, payment, bank contact, data-use action or spend. Dated source statements are evidence leads, not permission or final applicable law.

Evidence | Six written sources, unequal authority

Selected sources refreshed 13 November 2026. Government/operator policy sources differ from VC conceptual essays. Not independent empirical proof of a network effect.

NPCI overview	https://www.npci.org.in/product/upi/roles-responsibilities - returned NPCI operator responsibilities only; not a full TPAP/bank role definition or fee schedule. https://www.npci.org.in/circulars/upi - directory lists OC237/238 bank-contact pointers, not underlying pricing terms.
NPCI consolidated	https://www.npci.org.in/uploads/UPI_Consolidated_Circular_0641e910f7.pdf - dated 18 September 2026/version 1.0, selected scope, overview, participation, authentication and merchant clauses read. Fetch stops within AutoPay section; not full PDF/annexures/diagrams. Clause 1.5 retains pricing instruments.
Government summary	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2310586&lang=1&reg=48 - 15 September 2026 written summary read. Scoped stated framework, not operative rule verification or a fee quote. No current concentration-cap conclusion.
a16z, 13 Dec 2018	https://a16z.com/the-dynamics-of-network-effects/ - written article read. Saturation, heterogeneous participation, overlap and multihoming mechanisms; historical examples/cutoffs not universal laws or current company outcomes. VC perspective, not independent experiment.
NFX, Jun 2021	https://www.nfx.com/post/network-effects-manual - selected opening definition/taxonomy only, not entire manual/map/linked course. Strongest-defensibility and 70%-of-value claims not adopted or independently validated. Types do not prove benefits or ownership.
Authored evidence	Only controlled authored formulas and paper assumptions establish the numbers. No source/AV completeness, independent learner test, Excel/Sheets parity, real account/legal clearance or overall 10/10. Actual records stay blank.

Actual work | Not simulated mastery

Record only performed permission-cleared work; no real payment identifiers or private financial data.

Date/version	Role distinction	DiD and objection	Unit/margin correction	Policy missing scope	Reviewer finding	Delayed transfer

Actual work remains blank. Record only what occurred: dates, reasoning, error/source/correction, strongest objection and a new delayed transfer. No invented learner results, sponsor acceptance, actual budget release or signed rights.

Failure log | Authored execution only

Observed native tests are added only after execution. Not actual learner mistakes, independent human audit, Excel/Sheets parity, real authority or overall 10/10.

Controlled test	Observed authored outcome
Independent baseline arithmetic	PASS: 21 baselineformulas independently recomputed.
Changed completion scenarios	PASS: 6 changed completion sets including zero baseline/negative lift.
Changed unit-economics inputs	PASS: 10 changed economics sets including zero/negative margin.
Changed time/fee assumptions	PASS: 8 changed time/administration/provider-fee sets.
Invalid controls and isolation	PASS: 16 invalid controls and unaffected model isolation.
Broken-reference explicit holds	PASS: 6 broken references produce explicit calculation errors.
Restored baseline and clean caches	PASS: 24 changed sets + restored baseline; clean saved caches.
Actual/reviewer/transfer blank	PASS: actual learner/reviewer/delayed transfer remain blank.

Only the original synthetic course exercise at https://top1os.dev/ supplies the case figures. No new empirical reading, real participants, company records, Indian legal clearance, actual commitments or source completeness claimed.